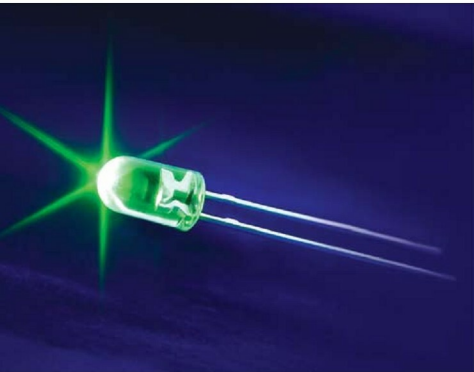


INTELLIGENT LIGHTING

Small, low-cost microcontrollers can be used to implement various types of electronic controls in lighting power-supply systems. Using light-emitting diode (LED) and fluorescent lighting as examples, here we show you how a microcontroller enables control and efficiency, as well as intelligence for closed-loop monitoring and communication



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Electronic lighting control is an emerging application in power electronics. One common driver of this market is the resulting efficiency. The addition of a small, low-cost microcontroller unit (MCU) increases efficiency besides providing other benefits in lighting power-supply systems.

For example, an MCU can be added to a LED lighting system for efficiently controlling the brightness

of the LED over its entire operating range. (LEDs provide a durable source of light with efficiencies exceeding those of incandescent bulbs.) Similarly, for fluorescent bulbs, which are also a very efficient source of light, electronic control via an MCU allows the bulb to dim to any level, thereby increasing efficiency. The MCU can also monitor bulb currents to ensure that its brightness level remains constant.

Other benefits of using an MCU are active power factor correction (PFC) to further increase the efficiency, battery

charging for portable lighting applications and the opportunity to integrate communication protocols such as DALI and DMX512.

The energy used by lighting

A significant amount of energy usage can be saved by switching to more efficient types of light sources and the use of electronic controls. The state of California has some of the most strict energy regulations in the US, and for good reason. It uses approximately 265,000 GWh of electricity each year, and the peak demand increases annually at a rate of 2.4 per cent. This increase in demand is equivalent to the output of three 500MW power plants. It is estimated that their energy-reduction efforts have saved \$36 billion so far, and will save another \$43 billion by 2013.

The California Energy Commission performed a comprehensive study to document the lighting energy used by the state. The residential data was obtained from 700 homes, and the commercial data was obtained from 1500 commercial buildings. The commercial buildings were further separated into ten categories.

This study showed that residential lighting consumed 8 per cent of all the energy used by the state. Commercial lighting consumed 14 per cent of all the energy used by the state.

The efficiency of a light source, or efficacy, is defined as the amount of light output divided by the electrical input power. The light output